



WE BRING HIGH-TECH TO LIFE

> FEMKE STORM

> 12-05-2026



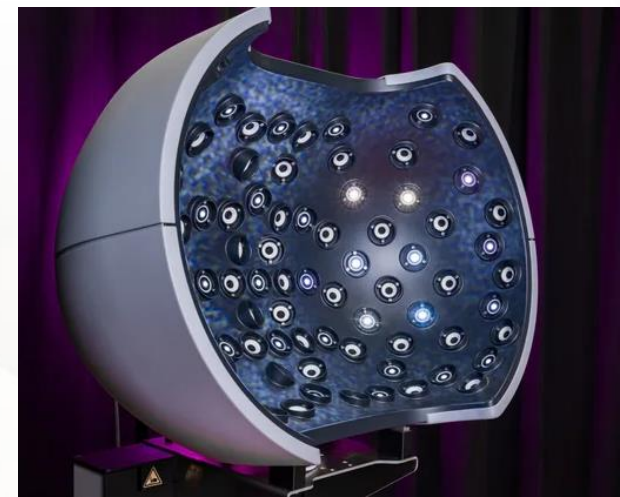
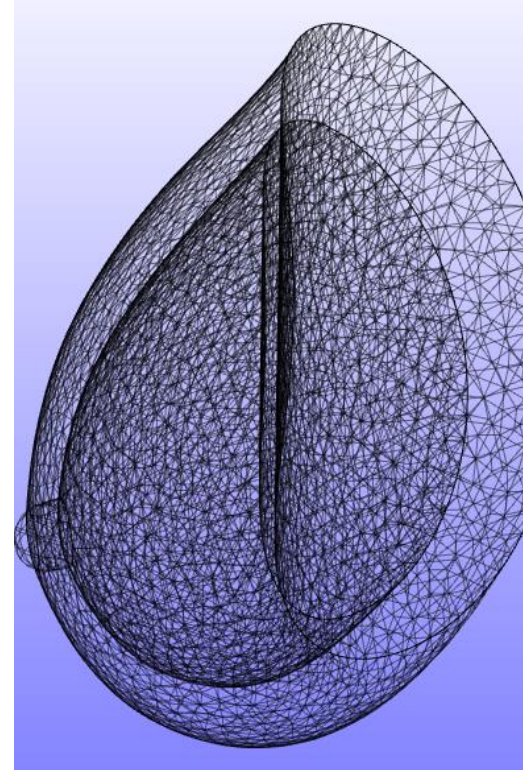
TISSUE MODELING FOR BREAST CANCER SCREENING

INTERNSHIP @ MATHWARE

AS PART OF MASTER BIOMEDICAL ENGINEERING @ TU/E

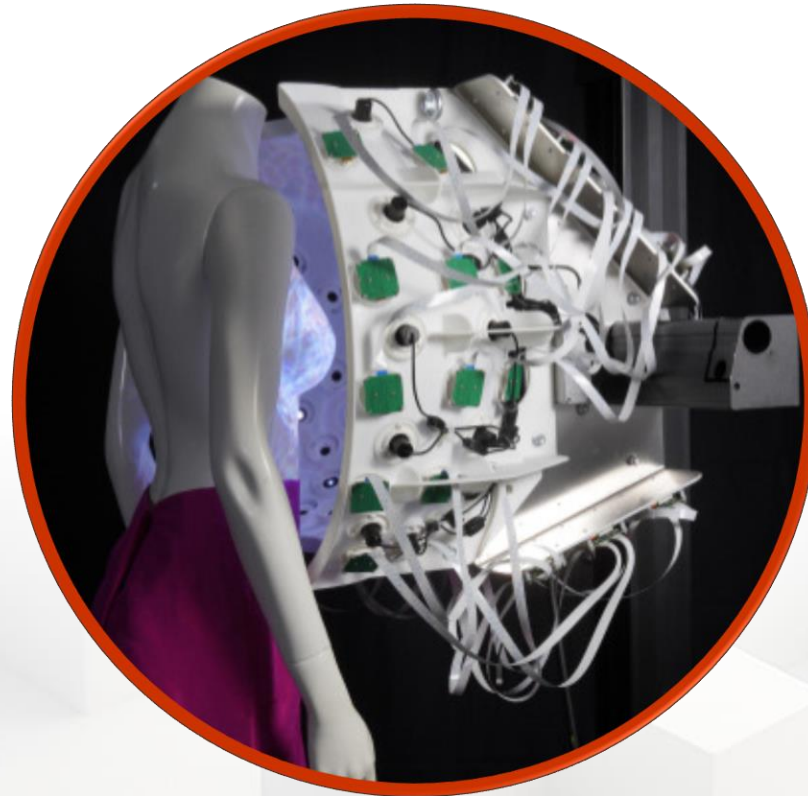
Femke Storm

12-05-2026

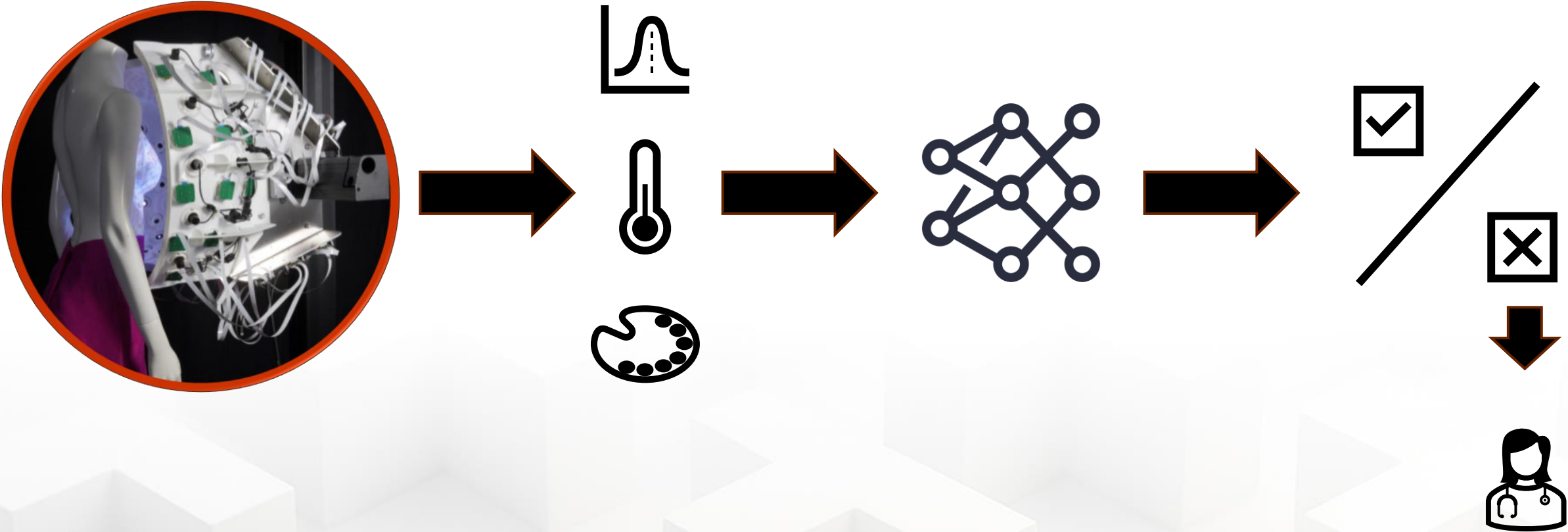


EARLY WARNING SCAN

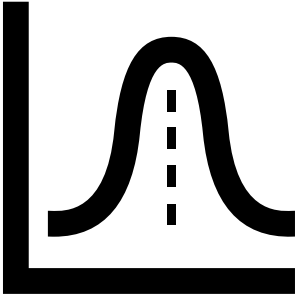
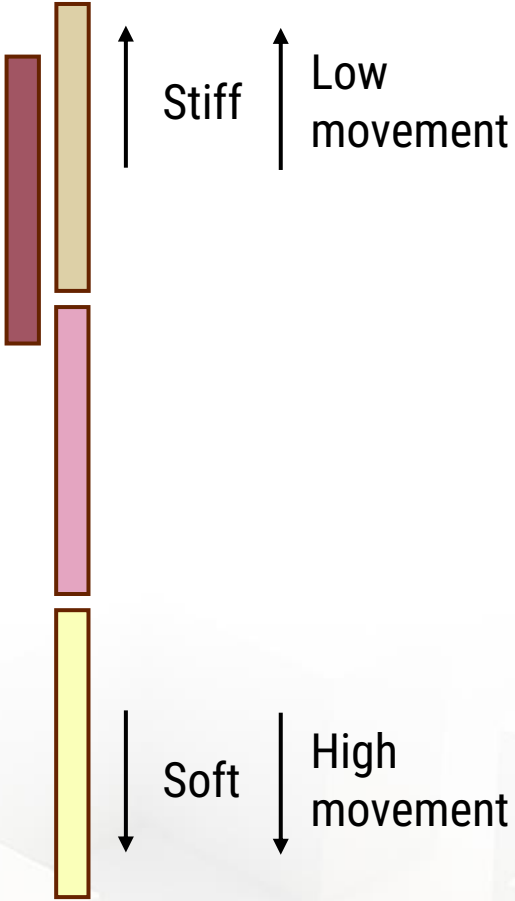
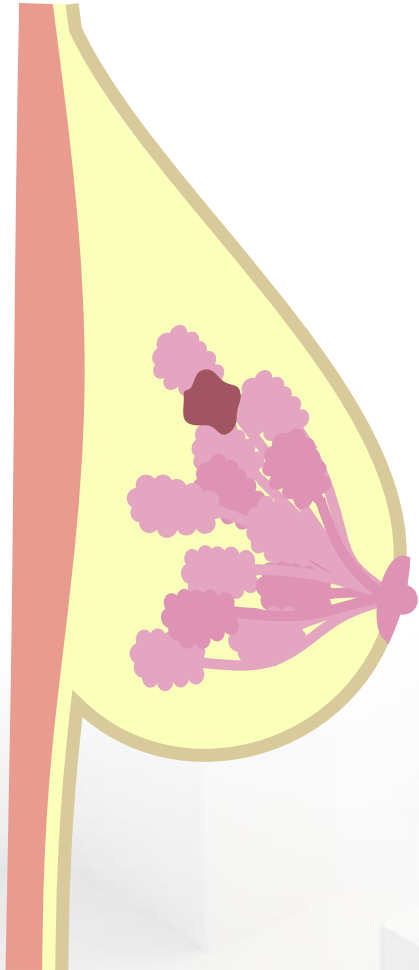
- Early detection of breast cancer
- Cheap, widely available, noninvasive alternative to routine mammography
- Alternative for people with dense breast tissue



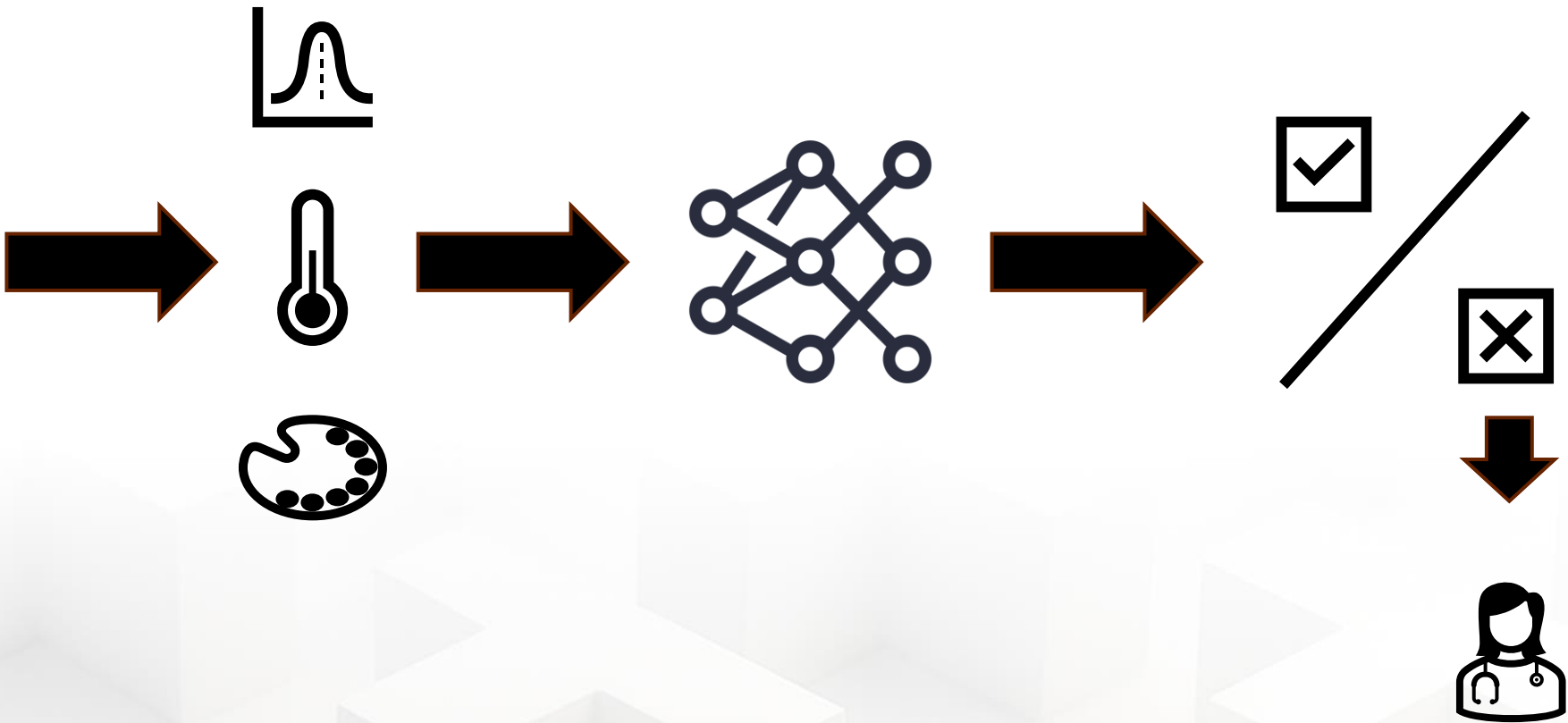
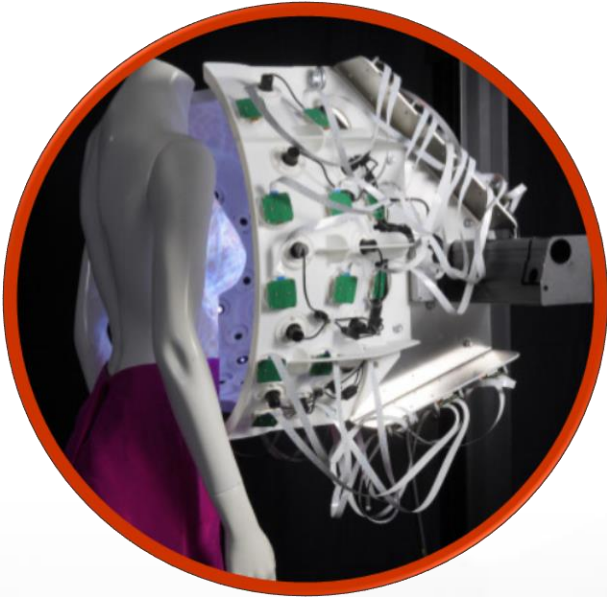
EARLY WARNING SCAN



BREAST MOVEMENT

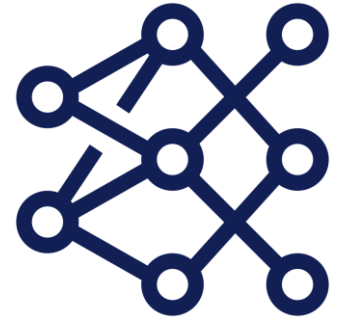


EARLY WARNING SCAN

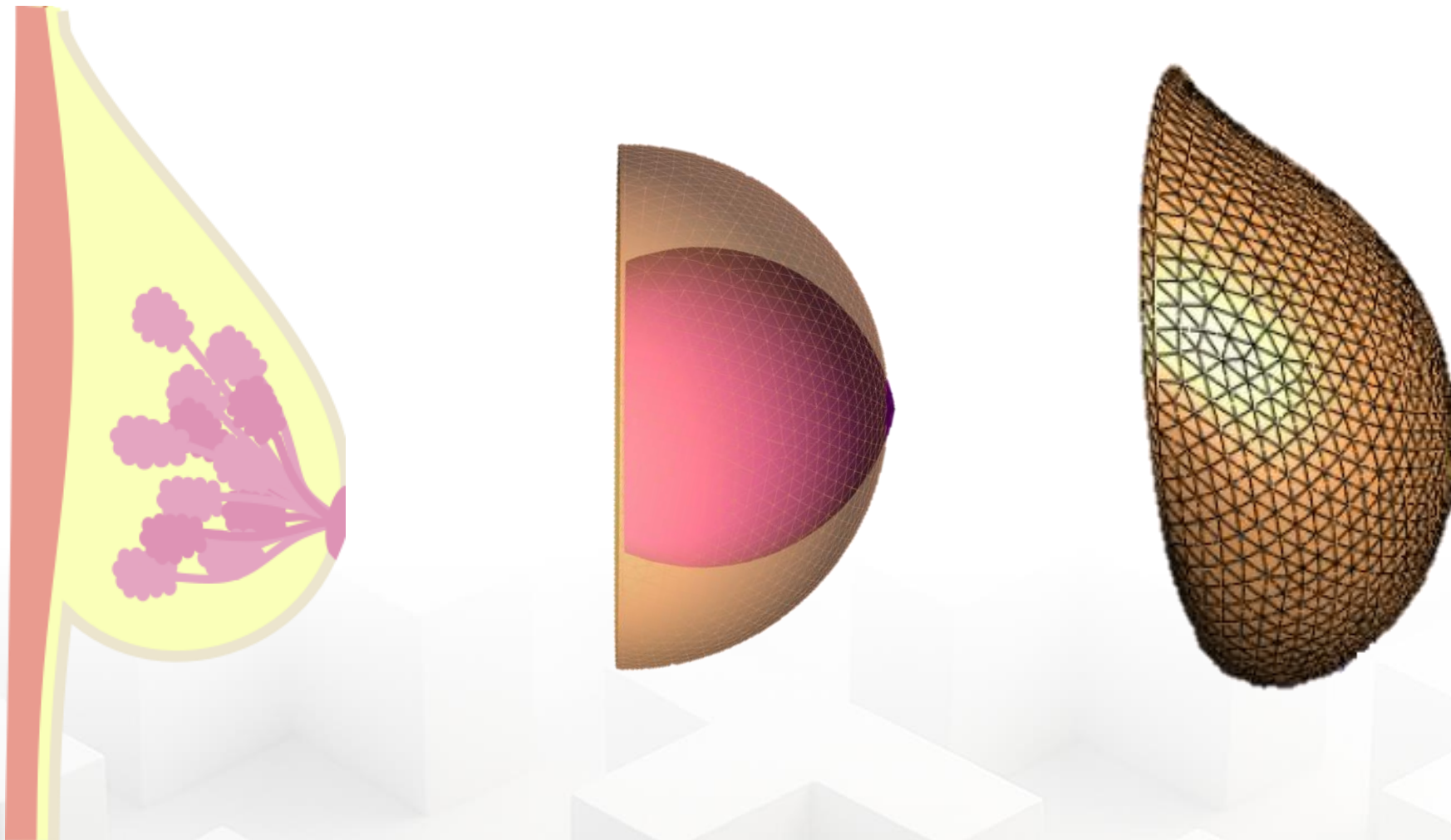


EARLY WARNING SCAN

- Needs training data
- Real-life training data difficult and expensive to obtain in large quantities
- Bulk training could be done on synthetic data



PREVIOUS WORK

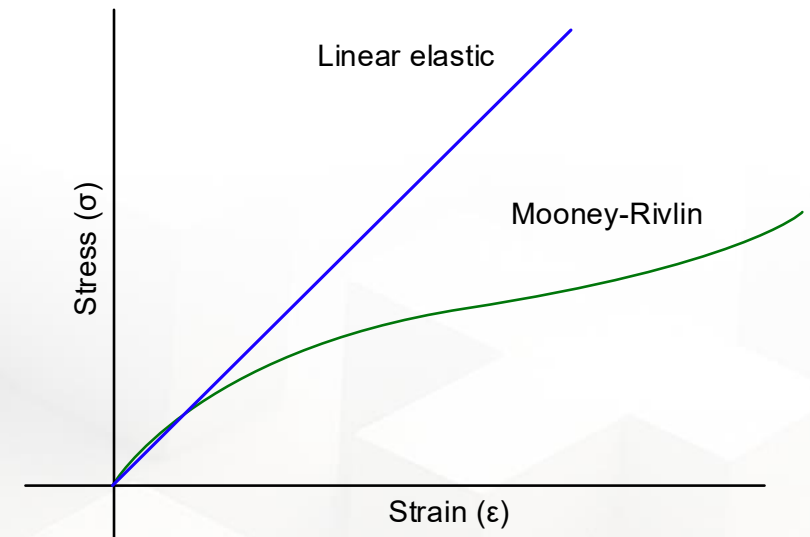
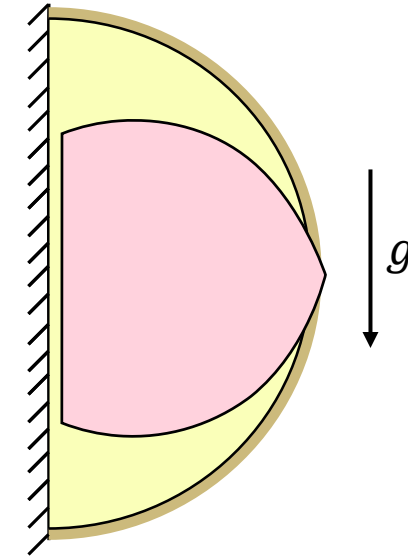


PREVIOUS WORK : FEM MODEL

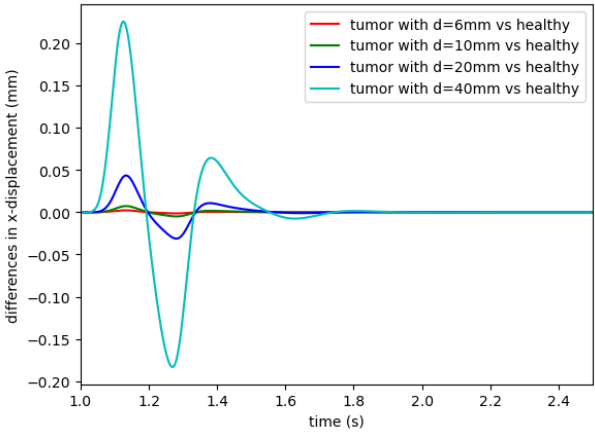
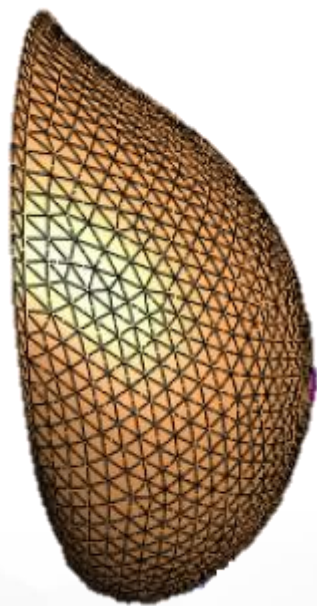
- $\nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} = 0$ on Ω
- $\mathbf{u} = 0$ on Γ_{chest}
- Material model Mooney-Rivlin
- FEBio: basic Mooney-Rivlin model with parameters C_1, C_2 and K

$$\Psi = \underbrace{C_1 (\tilde{I}_1 - 3) + C_2 (\tilde{I}_2 - 3)}_{\text{Deviatoric}} + \underbrace{\frac{1}{2} K (\ln J)^2}_{\text{Volumetric}}$$

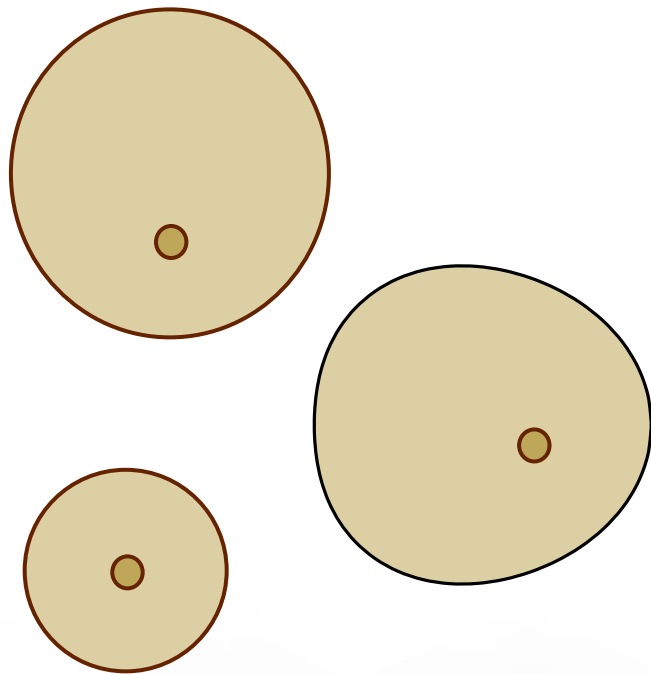
- Assumption: $C_1 \approx C_2 \approx \frac{1}{4} G_0$
- Ratio $G:K \sim 100$ to ~ 1000 for incompressible material



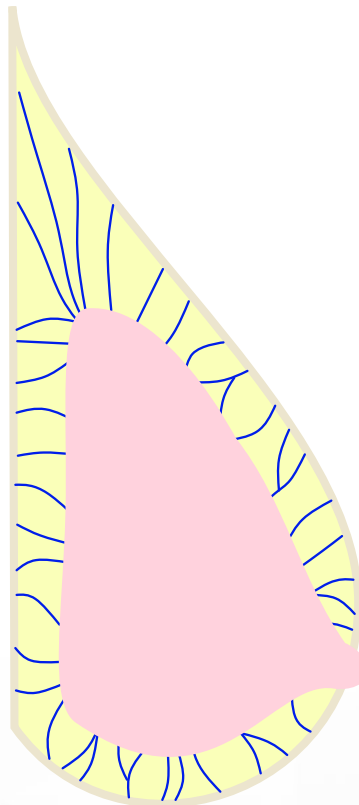
PREVIOUS WORK



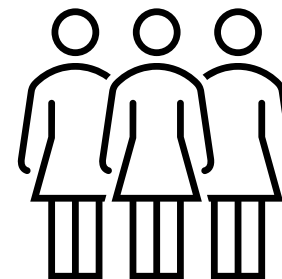
OBJECTIVES



➤ **VARIABLE BREAST GEOMETRY**



➤ **COOPER LIGAMENTS IN ADIPOSE TISSUE**

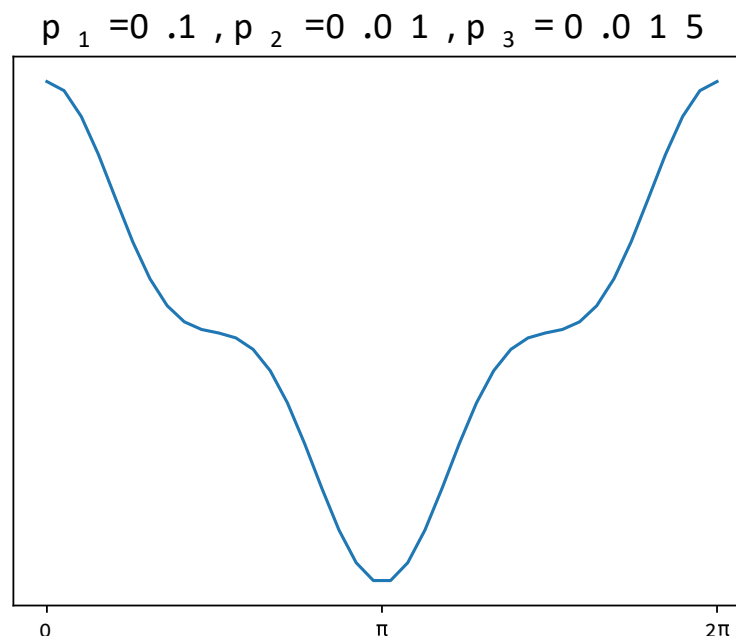
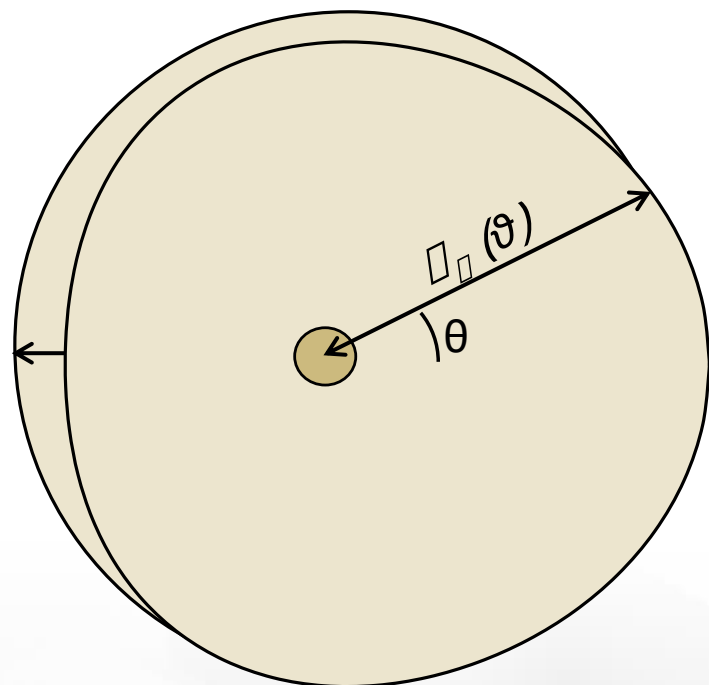


➤ **COMPARISON WITH REAL-LIFE DATA**

VARIABLE GEOMETRY

Developing a parametric geometry to more accurately describe the variation in breast shapes between individuals

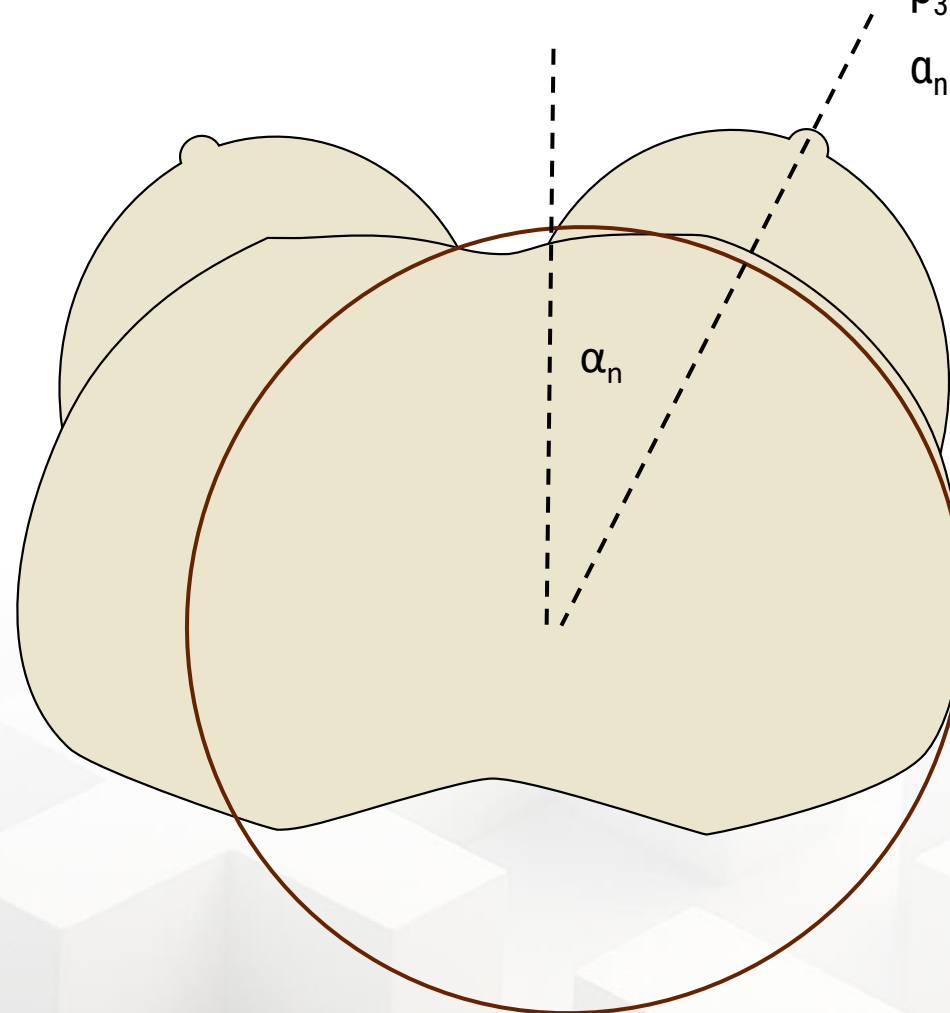
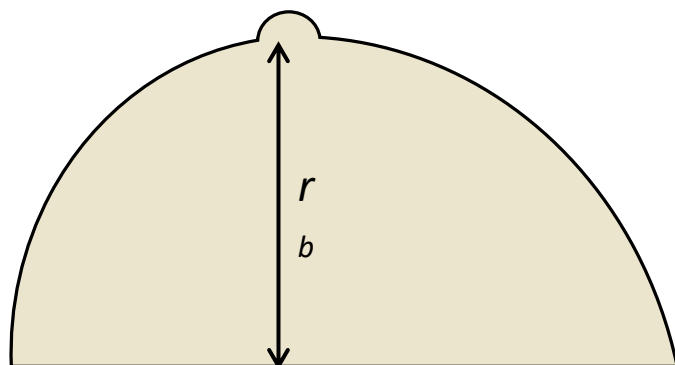
VARIABLE GEOMETRY



r_b Radius breast
 p_1 Asym. par. 1
 p_2 Asym. par. 2
 p_3 Asym. par. 3

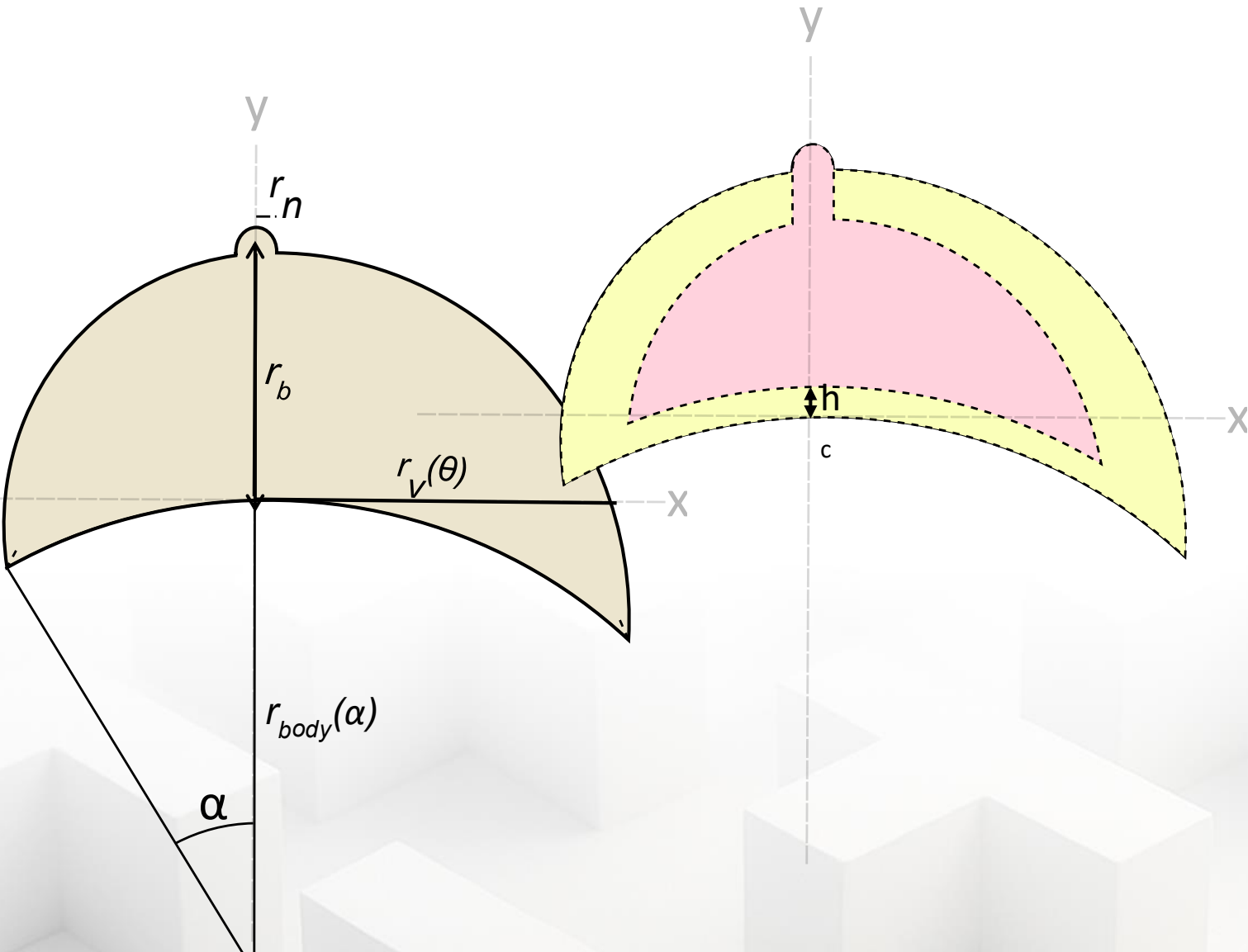
$$r_v = r_b \left(1 + \sum_{i=1}^3 p_i (\cos(i\theta) + 1) \right)$$

VARIABLE GEOMETRY



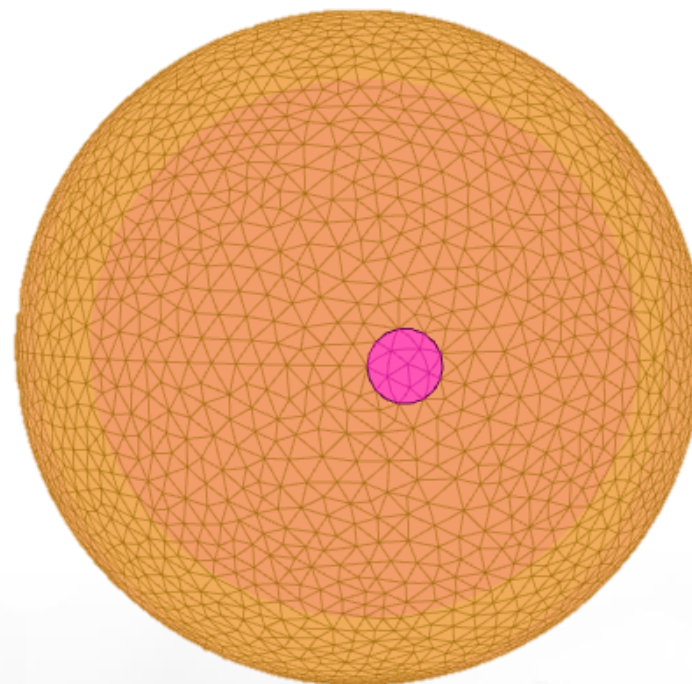
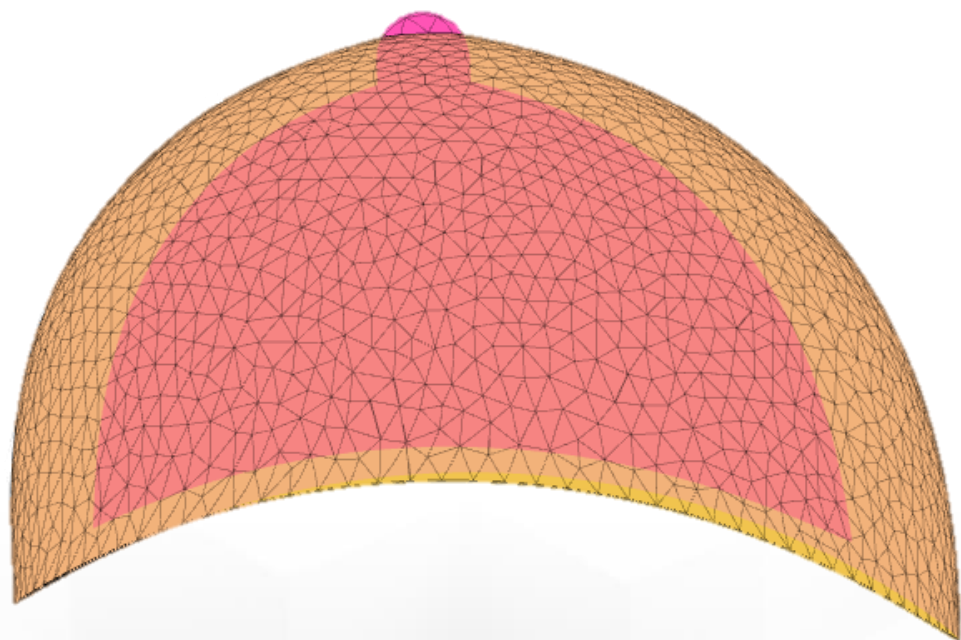
- r_b Radius breast
- p_1 Asym. par. 1
- p_2 Asym. par. 2
- p_3 Asym. par. 3
- α_n Angle nipple-chest

VARIABLE GEOMETRY

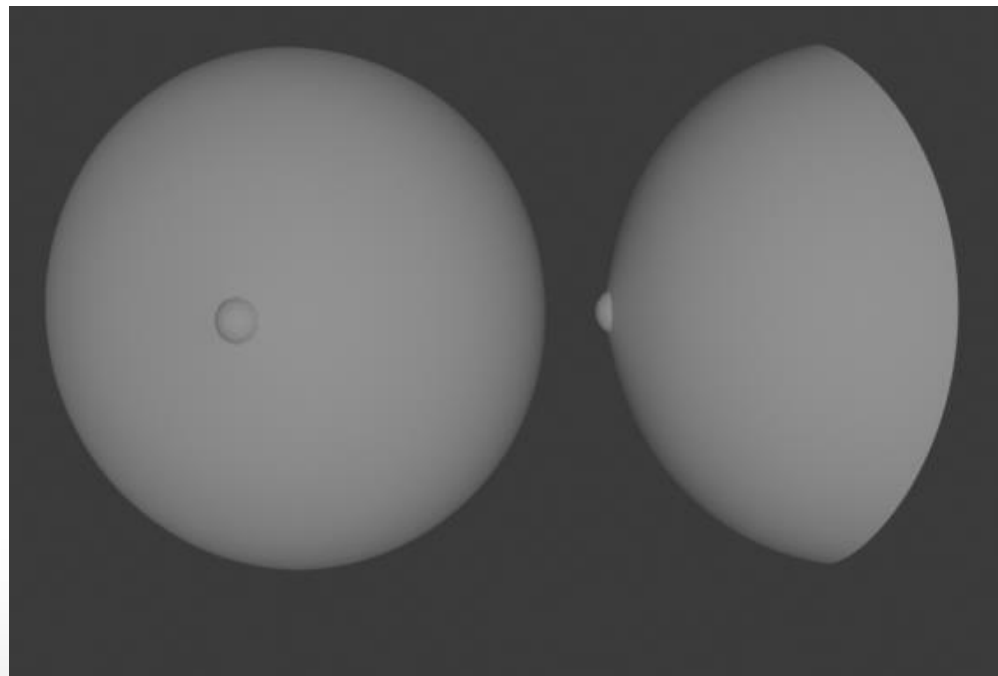


- r_b Radius breast
- p_1 Asym. par. 1
- p_2 Asym. par. 2
- p_3 Asym. par. 3
- α_n Angle nipple-chest
- s_y Scaling factor y-axis
- s_{xz} Scaling factor xz-axes
- h_c Thickness chest wall

RESULTING GEOMETRY



RESULTING GEOMETRY



COOPER LIGAMENTS

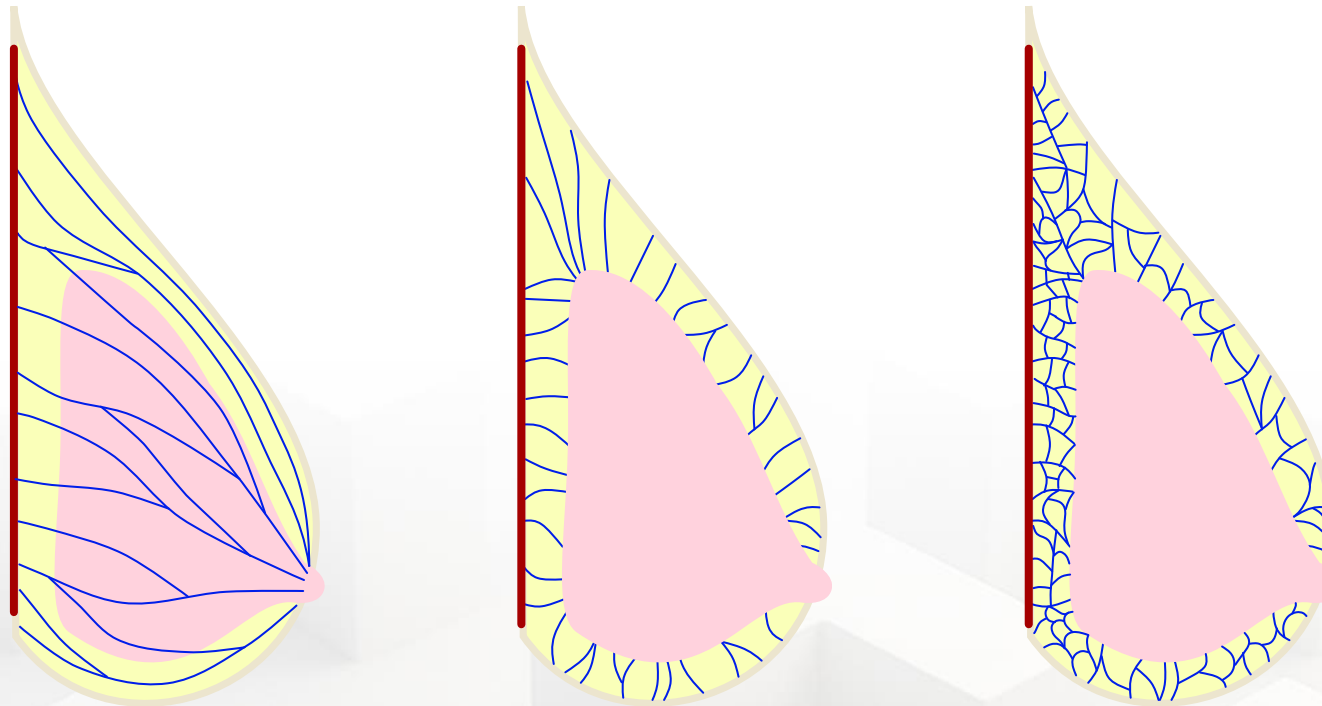
Adding additional shape-defining structures of the breast

COOPER LIGAMENTS

- Collagen fibers within adipose tissue between glandular tissue and skin
- Thought to have large impact on breast shape
 - With age: reduction of cooper ligaments causes change in breast shape
 - In breasts with tumors: can cause visible skin disturbances (dimpling, orange peel structure)
- Therefore relevant to have in the model!

DIRECTION OF COOPER LIGAMENTS

- Literature does not have consensus on exact direction/structure of Cooper Ligaments
- Either way; need for anisotropic material model



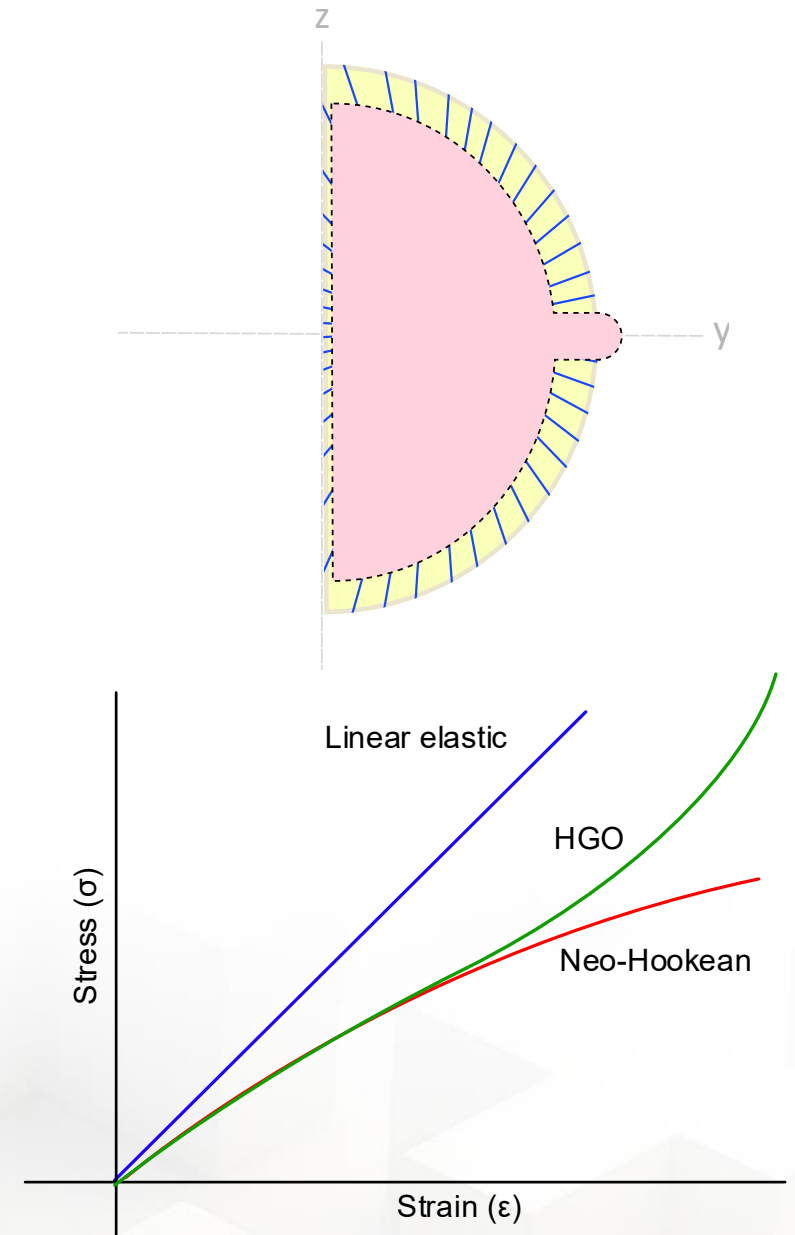
IMPLEMENTATION OF COOPER LIGAMENTS

- Holzapfel-Gasser-Ogden material model
 - Hyperelastic, anisotropic, fiber-reinforced material
 - One fiber family in one direction
 - Fiber direction spherical with center

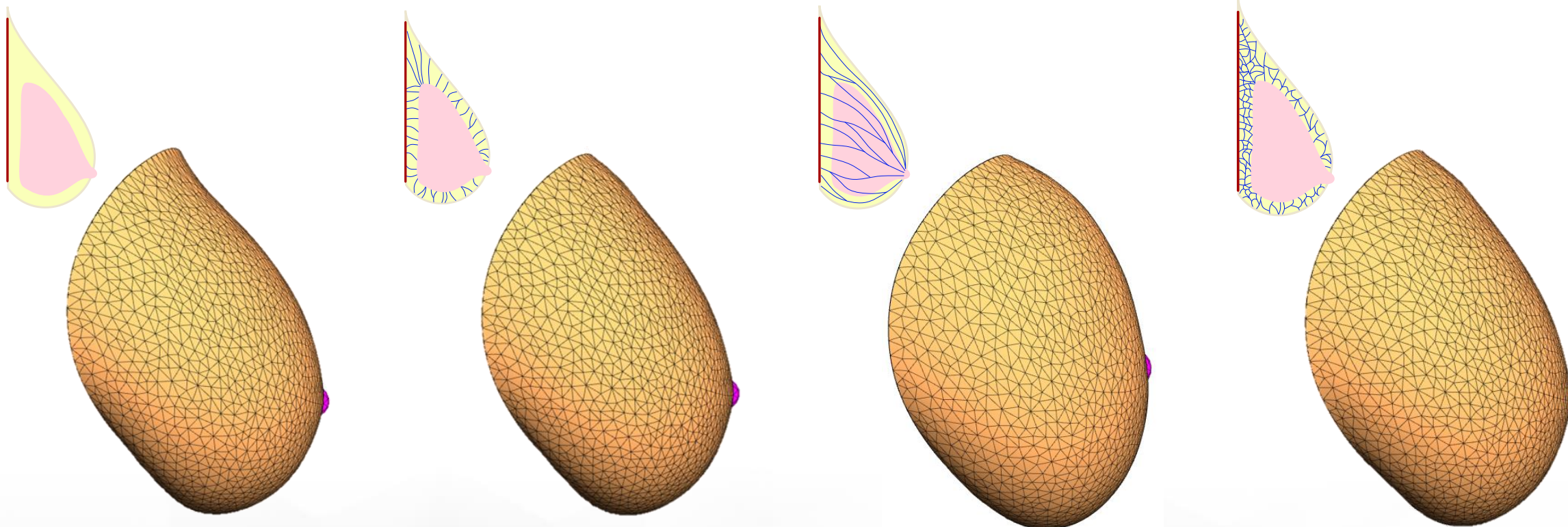
$$\Psi = \underbrace{\frac{c}{2}(\tilde{I}_1 - 3) + \frac{k_1}{k_2}(\exp(k_2\langle\tilde{E}_\alpha\rangle^2) - 1)}_{\text{Deviatoric}} + \underbrace{U(J)}_{\text{Volumetric}}$$

$$\tilde{E}_\alpha = \kappa(\tilde{I}_1 - 3) + (1 - 3\kappa)(\tilde{I}_{4\alpha} - 1)$$

- Material parameters
 - Assumption: $c \approx G_0$
 - Wide variation in values for k_1 and k_2 in literature



IMPLEMENTATION OF COOPER LIGAMENTS



No Cooper ligaments

$$c = 300 \text{ Pa}$$

$$k_1 = 0$$

With Cooper ligaments

$$c = 300 \text{ Pa}$$

$$k_1 = 5890 \text{ Pa}$$

$$k_2 = 12.5$$

With Cooper ligaments

$$c = 300 \text{ Pa}$$

$$k_1 = 5890 \text{ Pa}$$

$$k_2 = 12.5$$

With Cooper ligaments

$$c = 300 \text{ Pa}$$

$$k_1 = 5890 \text{ Pa}$$

$$k_2 = 12.5$$

$$\kappa = 1/6$$

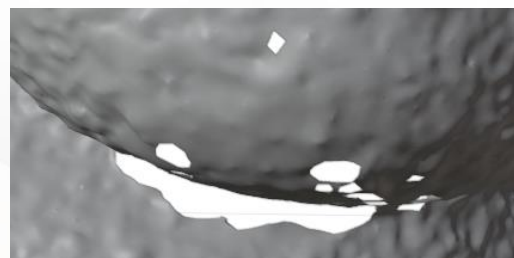
OBJECT COMPARISON

Defining a measure to compare model shape to real breast shapes

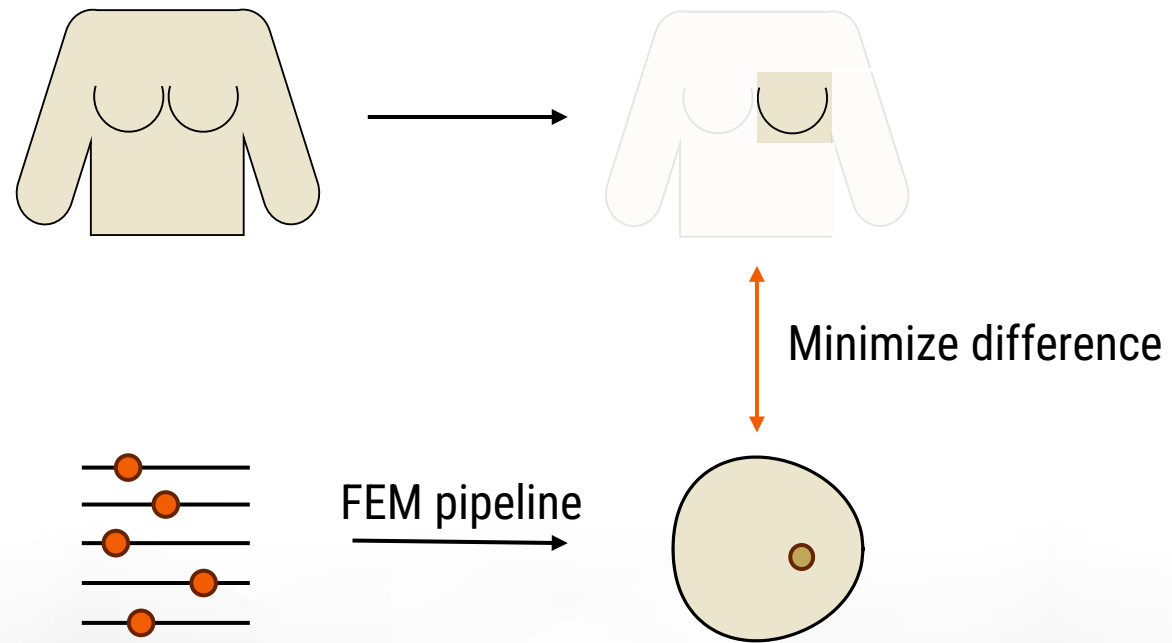
Optimizing geometry parameters to fit model output to real-life data

REAL-LIFE DATA

- 19x 3D scan of upper torso from Fontys
 - Both breasts, sometimes arms/neck/belly
- Lots of noise and gaps
- Static data only

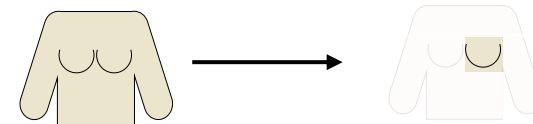


OBJECTIVE

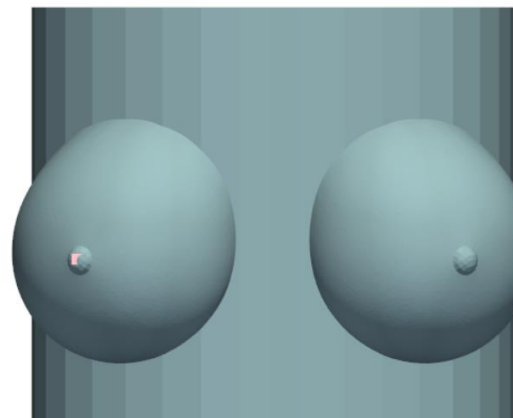


IMPORT DATA

- Import 3D scan
- Manually select nipple



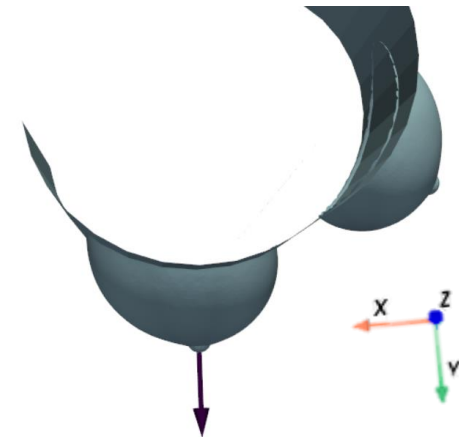
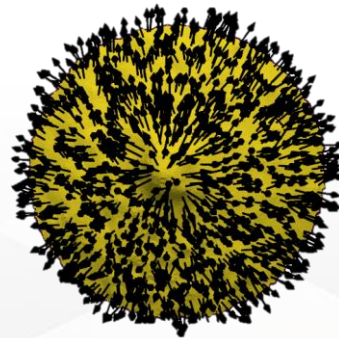
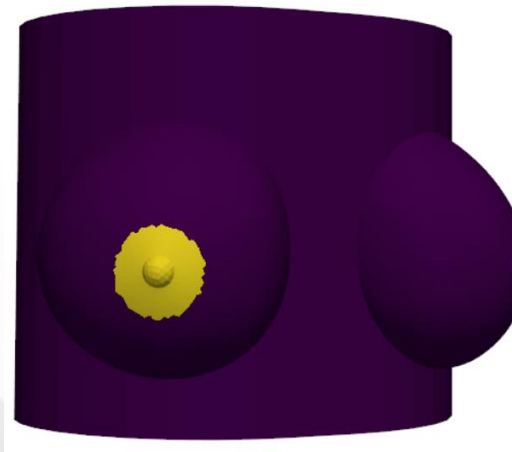
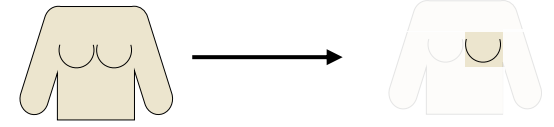
Click point for nipple. press enter to confirm, press Q when done.



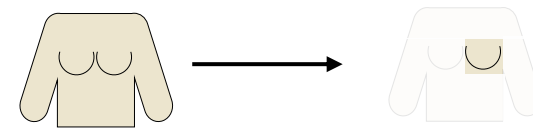
The original image was replaced with a synthetic torso due to the sensitive nature of the data

ORIENT BREAST

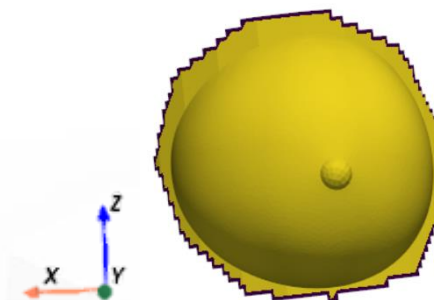
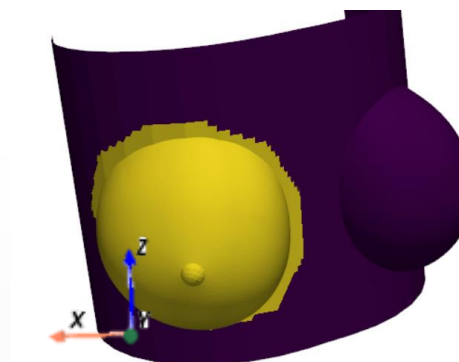
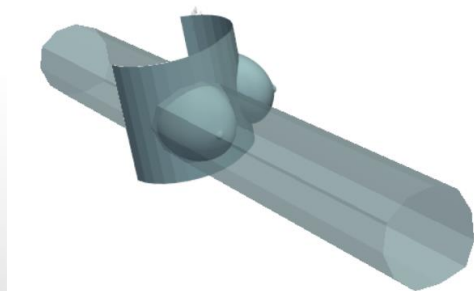
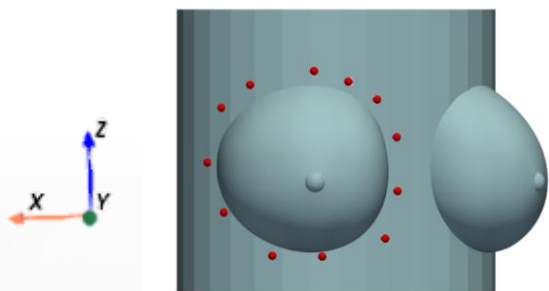
- Take all points within 2 cm from nipple
- Compute normals for all cells in that area
- Compute average normal
- Rotate and translate skin such that nipple normal is in yz-plane



SEGMENT BREAST FROM THE TORSO

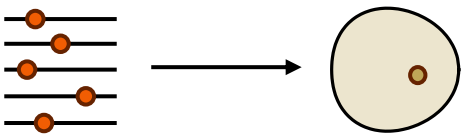


- Select points around the breast
- Create “cylinder” in the y-direction using those points
- Select all cells on skin surface within cylinder
- Remove all other cells



The original images were replaced with a synthetic torso due to the sensitive nature of the data

GENERATE FIRST MODEL



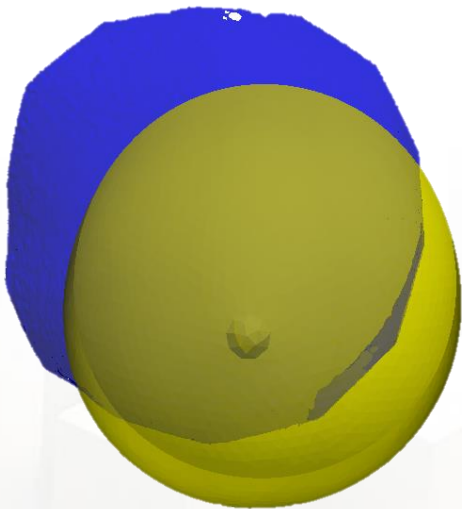
- Variable parameters:

r_b	Radius breast	0.07 m
p_1	Asym. par. 1	0
p_2	Asym. par. 2	0
p_3	Asym. par. 3	0
α_n	Angle nipple-chest	22.5

- Fixed parameters:

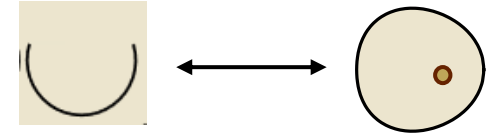
s_y	Scaling factor y-axis	0.8
s_{xz}	Scaling factor xz-axes	0.7
h_c	Thickness chest wall	0.005
	Material parameters	*

- Generate breast with initial parameters and align nipple



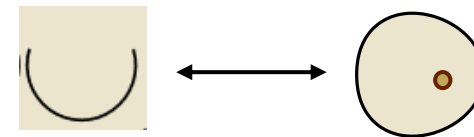
The original images were replaced with a synthetic torso due to the sensitive nature of the data

PARAMETER OPTIMIZATION ALGORITHM

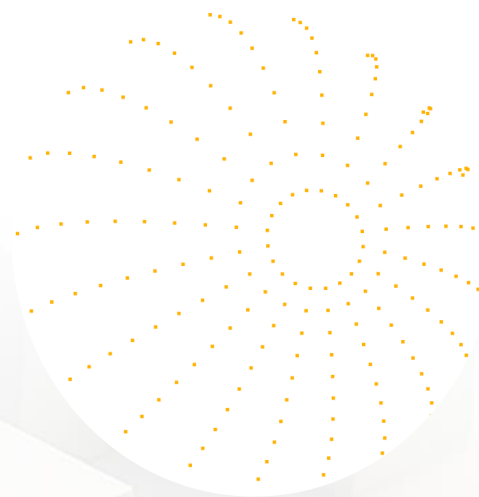
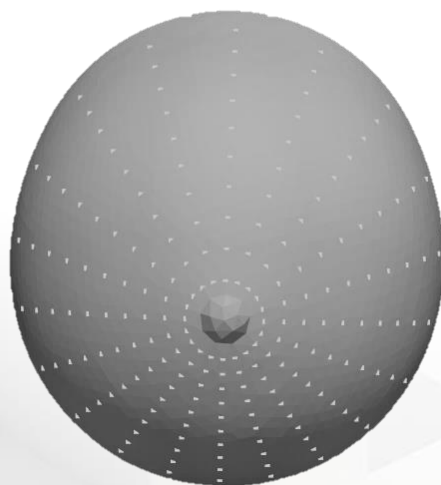


- Iterations are expensive
 - 3-25 minutes depending on model size
- Derivative-free optimization method -> LIMOLS
- Multiple residuals
 - Values can be both negative and positive
 - Fixed number

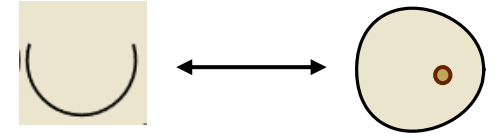
GENERATING LIMOLS RESIDUALS



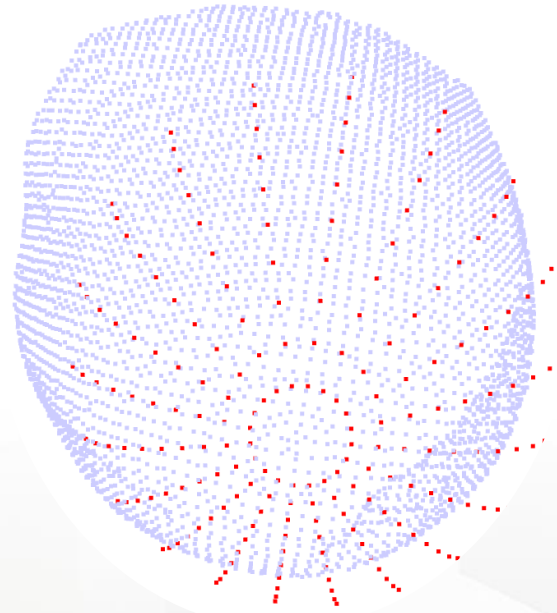
- Generate 200 points in XZ plane on 10 evenly distributed lines in breast surface projection
- Find y-coordinate for projection on breast surface



GENERATING LIMOLS RESIDUALS



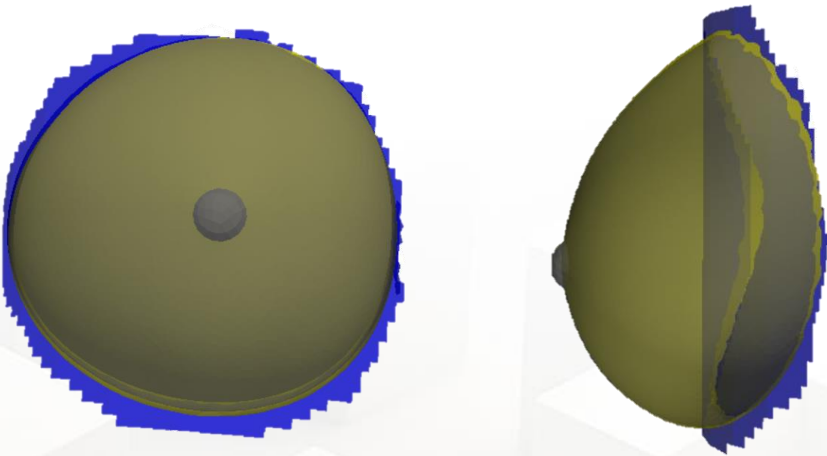
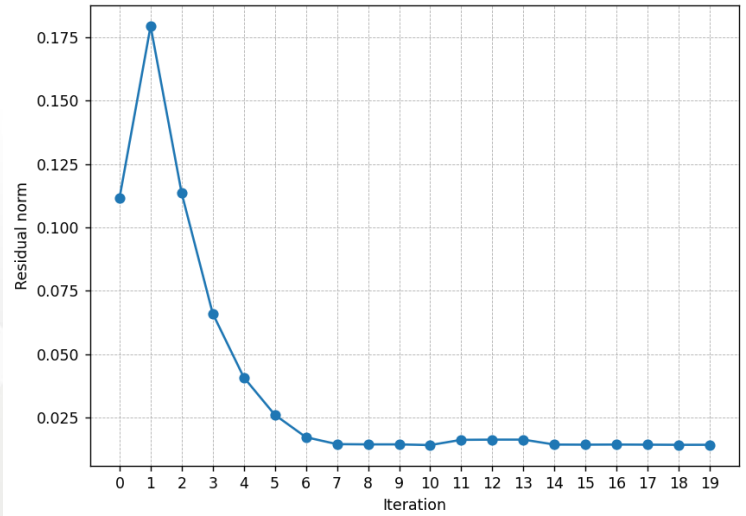
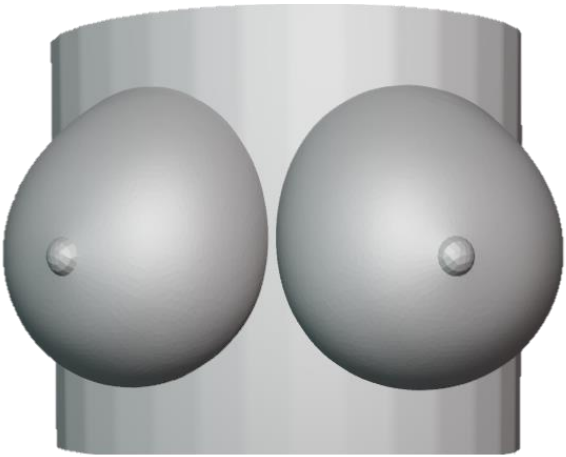
- For each projection point find closest point on target mesh using KD-Tree
- Return difference in 3 dimensions for all 200 points
 - -> input for LIMOLS



RESULTS

- Test case: synthetic torso with known ground truth

Parameter	Truth	Initial guess	Result
r_{breast}	0.048	0.07	0.0487
α_{nipple}	31.7	22.5	34.2



RESULTS

- Optimization results for real-life data not very accurate due to higher stiffness in model

The original images removed due to the sensitive nature of the data

DISCUSSION

- Shape of breast heavily dependent on material parameters
 - Large range between individuals
 - Hard to optimize to because of FEM/FEBio limitations
- Optimization to 1 image unreliable
 - Future: optimize to multiple images of same subject (e.g. different positions)

SUMMARY

- Variable geometry settings with 8 parameters created
 - Asymmetry, curved breast base
 - Glandular tissue shapes more variable
 - Improved nipple shape

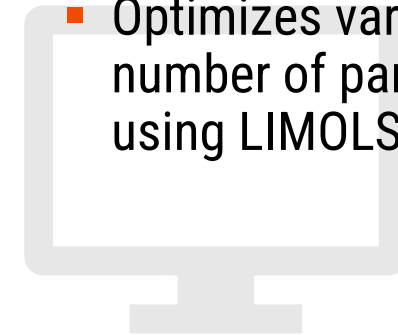
➤ VARIABLE BREAST GEOMETRY

- Added HGO material to adipose tissue for cooper ligament simulation



➤ COOPER LIGAMENTS IN ADIPOSE TISSUE

- Pipeline created for parameter optimization
 - Takes input of 3D scan
 - Manual segmentation of breast
 - Optimizes variable number of parameters using LIMOLS



➤ COMPARISON WITH REAL-LIFE DATA

FUTURE STEPS

- Improve FEM pipeline for lower stiffness (e.g. remeshing)
 - Then, look further into geometry optimization
- Explore optimization of material parameters
 - More/other data needed
- Improving and analyzing dynamic step
- Analysis of effect tumor in new geometry

The background features a complex arrangement of 3D rectangular blocks and L-shaped structures. These shapes are rendered in a gradient of colors, ranging from dark charcoal gray to a vibrant orange-red. The lighting creates strong highlights and shadows, emphasizing the three-dimensional nature of the objects. The overall composition is dense and abstract, with the shapes interlocking and overlapping in a way that suggests a complex, perhaps maze-like, structure.

QUESTIONS?